

IIITNR Minor Project Management Portal

Streamlining the Project Lifecycle for Students, Faculty, and Administrators.

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Core Objective



- A centralized web application built to automate group formation, project proposals, evaluations, and administrative oversight at IIT-Naya Raipur.
- Currently, managing minor projects across different branches (CSE, ECE, DSAI) involves a lot of manual coordination—forming groups, assigning faculty guides, tracking proposals, and conducting evaluations.
- Our portal solves this by providing a unified, role-based platform. Whether you are a student looking to form a cross-branch team, a faculty member managing multiple project groups, or an admin overseeing deadlines and limits, our application digitalizes and automates the entire workflow from semester start to final grading.

Technology Stack & Core Architecture



- **Frontend (User Interface):**
 - React.js + TypeScript (Built with Vite for fast performance)
 - Radix-UI (For accessible, clean, and minimalist components)
- **Backend (Server & Logic):**
 - Node.js with Express.js
- **Database (Data Storage):**
 - MongoDB (Storing Users, Groups, Projects, Proposals, and Evaluations)
- **Key Integrations:**
 - Authentication: Passwordless Email OTP (Institute emails only: @iiitnr.edu.in)
 - Real-time Features: Socket.io (For in-app chat, notifications, and progress updates)



Student Experience & Group Formation

- **The Student Journey (Flowchart/Steps):**

- **Onboarding:** Secure login via Institute Email (auto-detects branch/details).
- **Dashboard:** Personalized view of deadlines and current status.
- **Group Formation:**
 - Search for batchmates by name, roll number, or branch.
 - Cross-branch teams are fully supported (up to 3 members).
- **Surplus Pool:** Visibility of students without groups to ensure everyone finds a team.

- **Key Features:**

- Reversible groups (ability to dissolve a group before a proposal is approved).

Project Proposal & Mentor Allocation



- **The Proposal Process (Step-by-Step):**
 - **Drafting:** Group inputs Title, Objectives, Tags, & optional Attachments (e.g., PDFs).
 - **Faculty Selection:** Browse available faculty showing their current load (e.g., "5/7 groups").
 - **Submission:** Instant notification sent to the selected faculty member.
- **Faculty Dashboard & Review:**
 - **Review:** Faculty views proposal details and group member profiles.
 - **Action Flow:**
 -  Approve (1-click approval).
 -  Reject (Mandatory feedback required).
- **Capacity Controls:**
 - Strict limits enforced (Max 21 students or 7 groups per faculty).

Administrative Controls & Global Oversight



- **Key Benefits (The Impact):**
 - **Time-Saving:** Replaces messy email threads with a single, centralized workflow.
 - **Transparency:** Clear visibility into faculty availability, project status, and deadlines.
 - **Fairness:** Enforces equitable faculty loads across departments.
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- **Future Scope (What happens next):**
 - **Integrations:** Connecting with the institute's primary LMS and Google Calendar for seamless scheduling.
 - **External Tools:** Adding hooks for Plagiarism checkers or native GitHub integration for software projects.
 - **Scalability:** Multi-semester archiving to track project history over years.

Database description

Collection (Table)	Primary Purpose	Key Fields Stored	Why It’s Crucial for the System
Users	Authentication & Identity Management	email (Institute only), role (Student/Faculty/Admin), profile details (Name, Branch, Roll No.)	Acts as the single source of truth for Role-Based Access Control (RBAC) and determines which dashboard the user sees.
Groups	Managing Student Teams	group_id, members (Array of max 3 User IDs), status (Pending/Locked), project_ref	Enforces the "max 3 students" rule and enables the "Surplus Students" feature by quickly identifying students without a group.
Projects & Proposals	Project Tracking & Faculty Matching	title, objective, tags, faculty_guide_ref (Faculty User ID), status, group_ref	Enforces faculty limits. The system queries this to ensure a faculty member stays under the strict 7 group / 21 student cap before allowing new approvals.
Evaluations	Grading & Administrative Oversight	project_ref, student_ref, marks (Numerical scores based on rubrics), evaluation_type (Mid/End term)	Separates raw scores from final grades, allowing admins to apply Grading Methods (Absolute, Relative, Manual) before officially releasing results.
Messages & Notifications	Real-Time Communication Layer	trigger_event (e.g., Deadline approaching), chat_logs, payload_data, timestamps	Tied directly to Socket.io to push instant alerts and chat updates to users without requiring a page refresh.

Thank You



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